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HTML QUICK STYLER

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ABSTRACT

This project presents an AI-assisted web page generator called HTML Quick Styler, developed using Python and the interactive UI framework Gradio. The system enables users to automatically generate modern and responsive HTML webpages by simply providing a page title, description, and selecting a preferred color style. The application integrates a large language model, IBM Granite 3.3 2B Instruct, accessed through the Hugging Face Transformers library, to support intelligent text generation and assist with dynamic content creation.

The architecture of the system consists of multiple functional modules, including a Color Palette Generator, CSS Generator, HTML Section Templates, and an HTML Page Builder. The Color Palette Generator provides predefined modern design themes such as modern, vibrant, ocean, forest, sunset, luxury, creative, and minimal. These palettes are applied automatically to produce consistent and visually appealing styles across the generated webpage.

The HTML Page Builder dynamically constructs the complete webpage by combining reusable section templates such as hero section, features section, about section, contact form, and footer. A CSS generation module creates responsive styles using modern layout techniques like CSS Grid and media queries to ensure compatibility across different screen sizes.

To improve usability, the system includes an interactive multi-tab interface built with Gradio. The interface provides five functional tabs: Generate Page, Live Preview, Color Palettes, Templates, and Guide. Users can instantly view the generated webpage in real time and copy the full HTML code for further customization or deployment.

The proposed system demonstrates how AI models and automated design templates can simplify the process of website creation for beginners, students, and developers. By reducing manual coding effort and providing ready-to-use responsive designs, the project highlights the potential of AI-driven tools in accelerating modern web development workflows.

1. INTRODUCTION

The HTML Quick Styler project is a web-based tool that automatically generates a complete HTML webpage with styling using predefined color palettes and templates. It integrates a user-friendly interface built with Gradio and an AI language model from IBM called IBM Granite to assist in generating text or content if needed.

The system allows users to enter a webpage title, description, and select a color style. Based on these inputs, the program generates a full HTML page with responsive CSS styling and multiple sections such as hero, features, about, contact, and footer.

2. OBJECTIVE

The main objectives of the project are:

- Automatically generate HTML web pages using Python.
- Provide multiple pre-designed color palettes for website styling.
- Create responsive CSS styles dynamically.
- Offer a simple interface for non-programmers to build web pages.
- Allow real-time preview of generated pages.

3. TECHNOLOGIES USED

Technology	Purpose
Python	Core programming language
Gradio	Web interface creation
PyTorch	Model execution
Hugging Face Transformers	Loading and using AI models
IBM Granite	AI text generation
HTML	Web page structure
CSS	Styling and layout

4. SYSTEM ARCHITECTURE

The system consists of the following main components:

1. Dependency Installation
2. AI Model Loader
3. Color Palette Generator
4. CSS Generator
5. HTML Section Templates
6. HTML Page Builder
7. Page Generation Function
8. Gradio User Interface

Workflow:

User Input → Color Palette Selection → HTML Builder → CSS Generator → Web Page Output → Preview

5. MODULE DESCRIPTION

5.1 Dependency Installation

The program first installs required libraries:

- gradio
- transformers
- accelerate
- torch

These libraries enable AI model loading and the web interface.

5.2 Import Libraries

```
import gradio as gr
from transformers import AutoModelForCausalLM, AutoTokenizer
import torch
```

These imports allow:

- Creating web interfaces
- Loading AI language models
- Running deep learning models

5.3 Loading the AI Model

The code loads the IBM Granite 3.3 2B Instruct model.

Main steps:

1. Define model name.

2. Load tokenizer.
3. Load the model with GPU acceleration if available.
4. Create a function to generate text responses.

Purpose:

- Generate AI text responses for prompts.

5.4 Color Palette Generator

This class stores eight predefined color themes.

Examples:

- Modern
- Vibrant
- Ocean
- Forest
- Sunset
- Luxury
- Creative
- Minimal

Each palette contains:

- Primary color
- Secondary color
- Accent color
- Background color
- Surface color

- Text color

The function get_palette() returns the selected color set.

5.5 CSS Generator

This function dynamically creates CSS styles based on the selected palette.

It defines:

- Root variables
- Page background
- Header styling
- Hero section
- Buttons
- Grid layout
- Cards
- Footer
- Responsive design

Example features:

- Sticky navigation
- Responsive sections
- Grid-based feature layout

5.6 HTML Section Templates

The program defines reusable templates for webpage sections.

Hero Section

Displays:

- Page title
- Introduction text
- Call-to-action button

Features Section

Shows three feature cards:

- Fast
- Responsive
- AI Powered

About Section

Displays information about the company or portfolio.

Contact Section

Includes a form with:

- Name field
- Email field
- Message box
- Submit button

Footer

Displays copyright information.

5.7 HTML Page Builder

The HTMLPageBuilder class assembles the full webpage.

Steps:

1. Generate CSS using selected palette.
2. Insert CSS into <style> tag.
3. Add header and all sections.
4. Return final HTML document.

Output includes:

- Complete HTML structure
- Embedded CSS styling
- Pre-built sections
- **5.8 Page Generation Function**

This function connects all modules.

Process:

1. Get palette from generator.
2. Send palette to page builder.
3. Build full HTML page.
4. Return generated HTML.

5.9 Gradio Interface

The interface is built using Gradio Blocks.

The application contains five tabs:

1. Generate Page

User inputs:

- Page title
- Page description
- Color palette

Click Generate HTML to create webpage.

2. Live Preview

Displays rendered HTML page.

3. Color Palettes

Displays information about available color themes.

4. Templates

Lists available webpage templates.

Provides instructions for using the tool.

6. PROGRAM WORKFLOW

1. User opens interface.
2. User enters page title and description.
3. User selects color palette.
4. System generates CSS from palette.
5. HTML builder combines sections.
6. Generated HTML is displayed as:
    - Code
    - Live preview.

7. FEATURES OF THE SYSTEM

- Automatic HTML page generation
- Multiple modern color themes
- Responsive CSS design
- Live preview functionality
- Simple user interface
- AI model integration

8. ADVANTAGES

- Reduces web development time
- Easy to use for beginners
- Customizable styling
- Works directly in browser
- No manual coding required

9. LIMITATIONS

- Limited page templates
- Static HTML output
- AI text generation not fully integrated into sections
- Requires internet to load AI model

10. CONCLUSION

The HTML Quick Styler project demonstrates how Python, AI models, and web interface frameworks can be combined to automatically generate web pages. By using predefined templates and dynamic CSS generation, the system simplifies web development and enables users to create styled webpages quickly without writing extensive code.